

Scientific Calculator

Project Presentation

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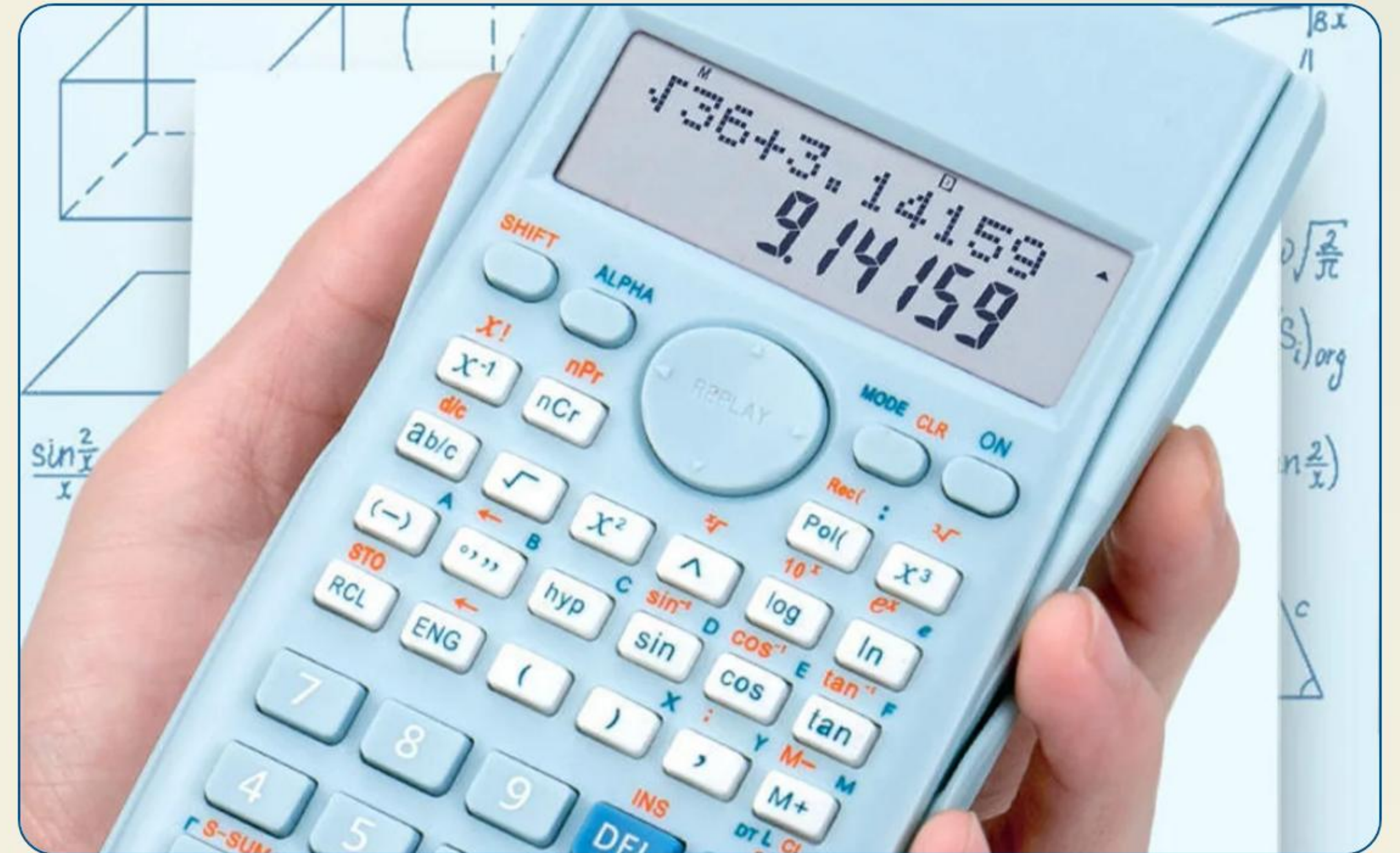
Course: CSE115 | Faculty: MSRB

System Overview

Computational Engine

A comprehensive C-based application that extends standard arithmetic to handle complex engineering calculations. Designed with modularity to ensure high precision and scalability.

- ✓ Real-time result processing
- ✓ Robust mathematical libraries
- ✓ High-precision floating point



The Development Team



Shafayet & Sabbir

Core Logic & Mathematical Algorithm
Implementation



Samin Yasar

System Architecture & Function
Modularity



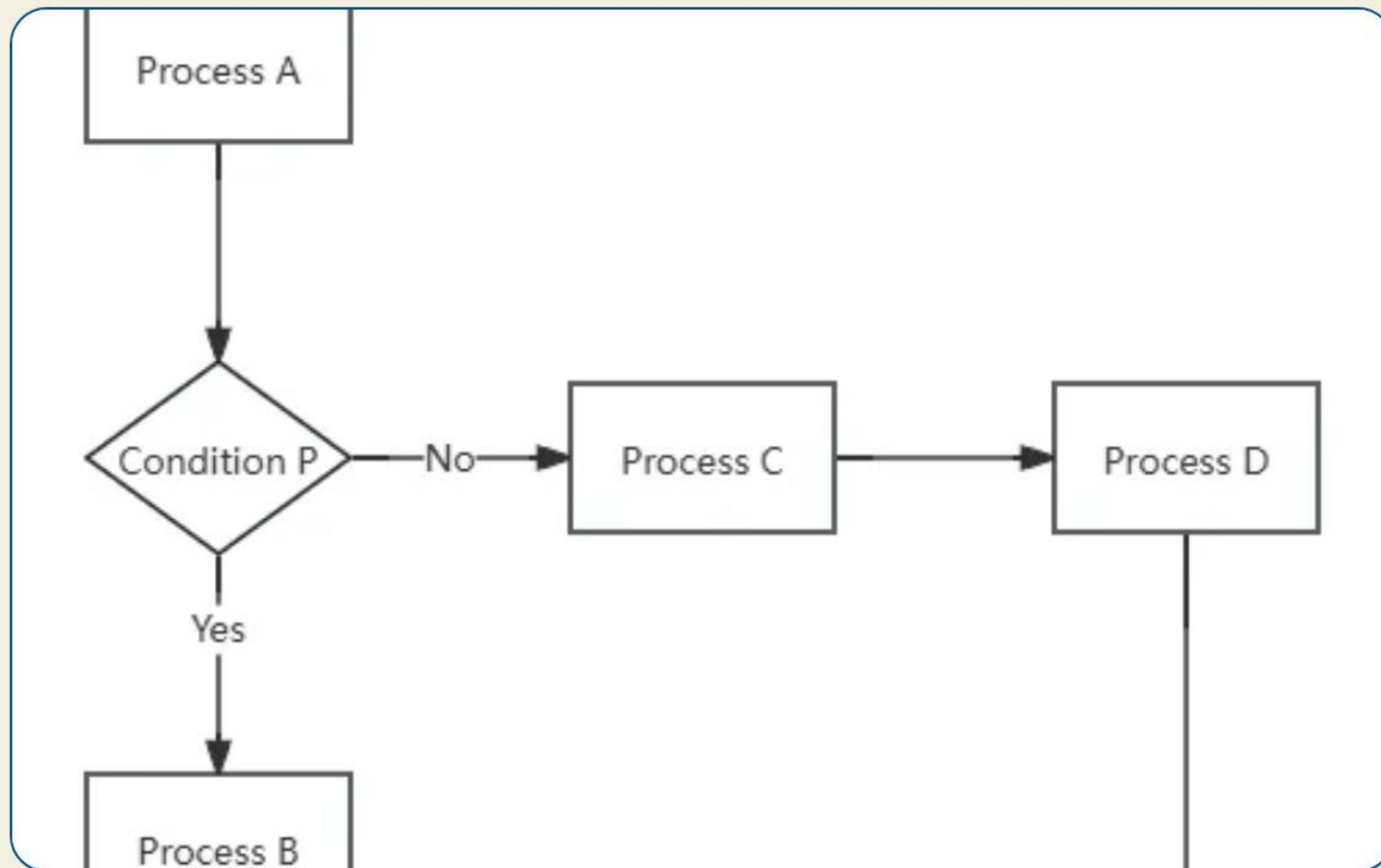
Samiul Islam

Error Handling & Quality Assurance
Testing

Core Library:

Category	C Function	Mathematical Application
Logarithmic	<code>log10(x)</code> , <code>log(x)</code>	Base-10 and Natural Logarithms
Trigonometric	<code>sin()</code> , <code>cos()</code> , <code>tan()</code>	Periodic wave and geometry calcs
Power/Root	<code>pow(x,y)</code> , <code>sqrt(x)</code>	Exponentiation and Radicals
Rounding	<code>ceil()</code> , <code>floor()</code>	Integer boundary mapping

Switch-Case Architecture



Efficient Selection

The program uses a switch-case engine to map integer inputs to specific function pointers, reducing overhead compared to nested if-else structures.

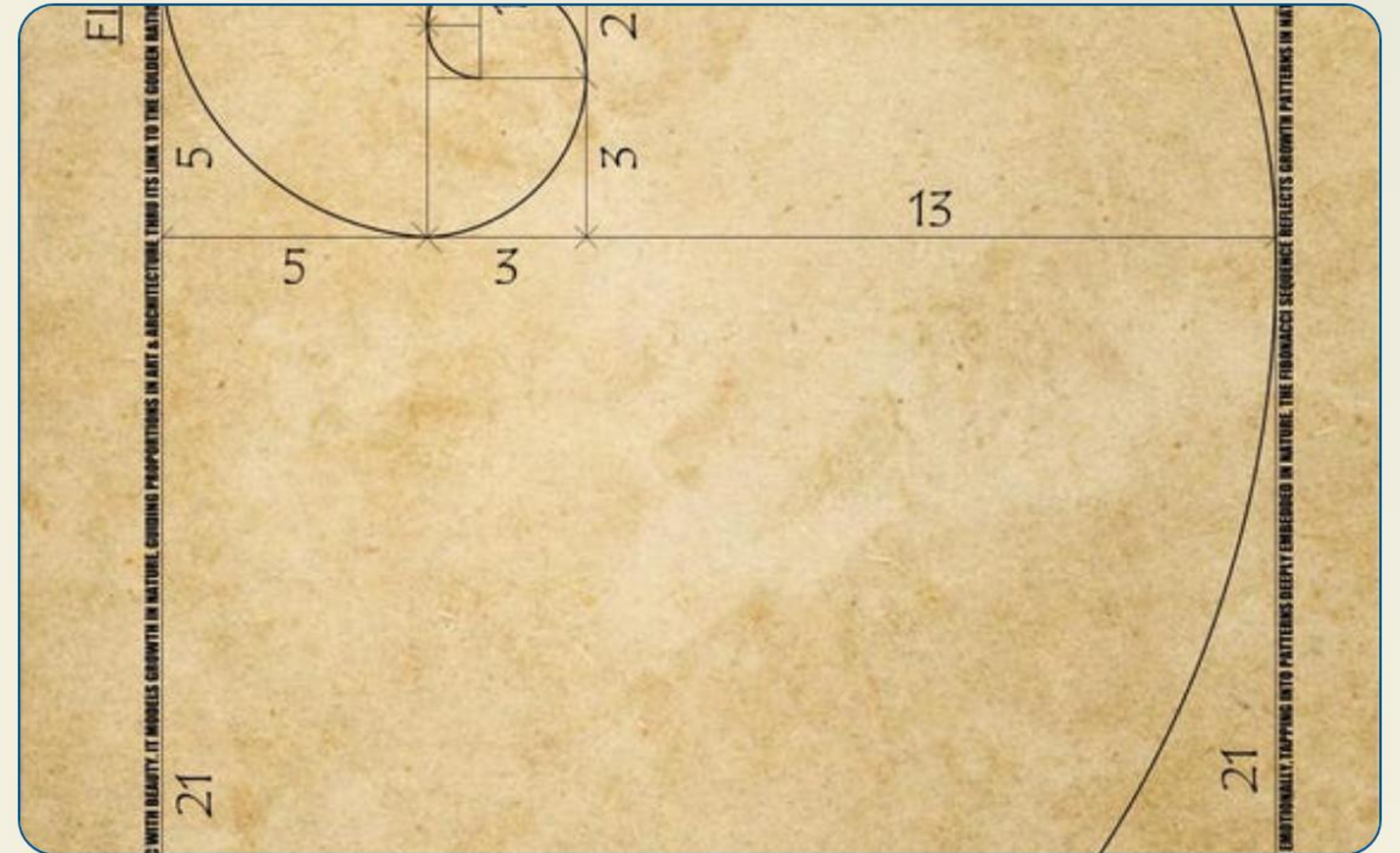
$O(1)$ complexity for operator selection

Advanced Trig Logic

Radian Conversion

C standard functions require radian inputs. Our calculator handles degree-to-radian conversion internally.

$$\text{rad} = \text{deg} \times \frac{\pi}{180}$$



Fault Tolerance



Divide by Zero

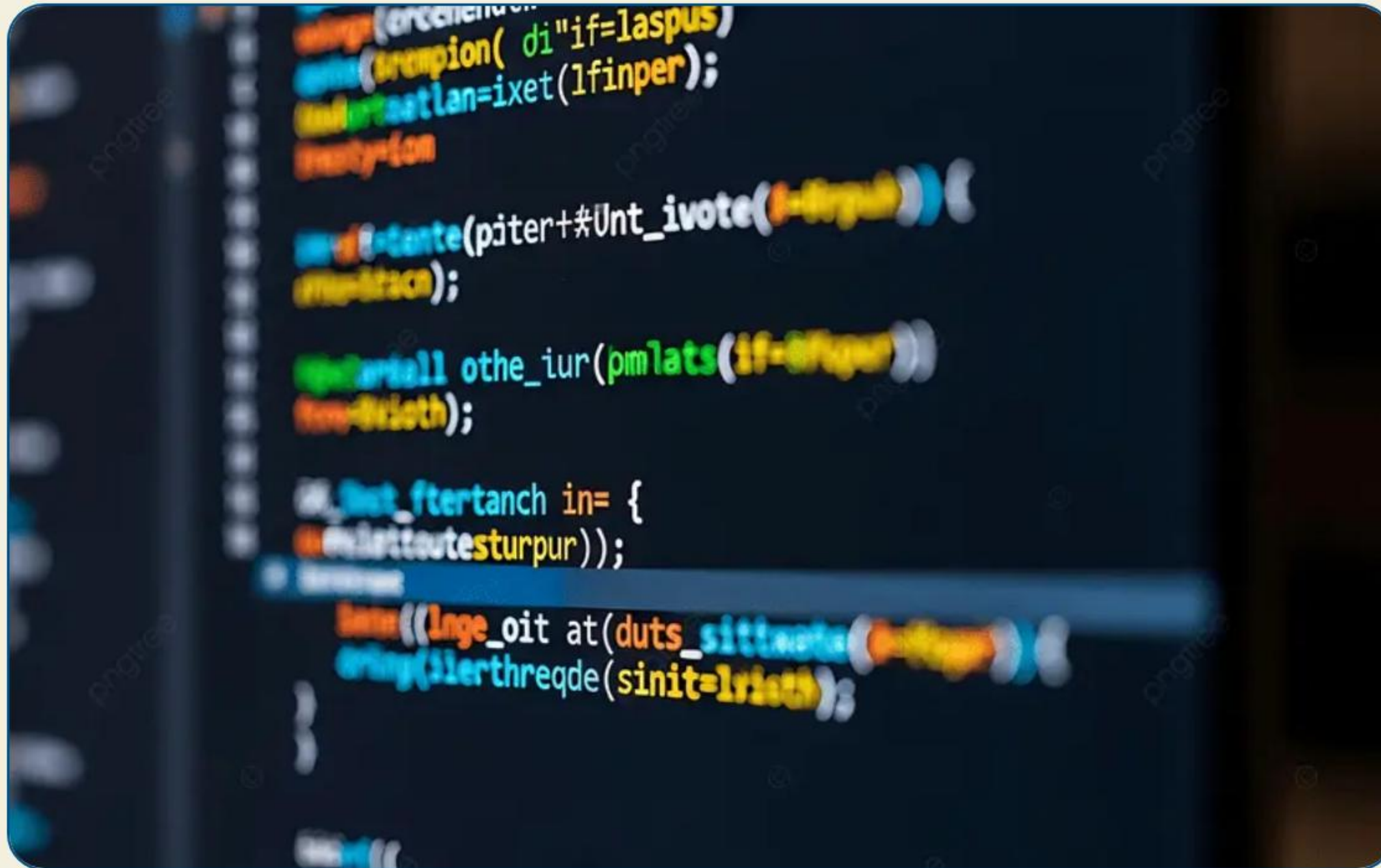
Strict checks on divisors to prevent program crashes during basic arithmetic.



Domain Errors

Handling square roots of negative numbers and logarithmic domain violations.

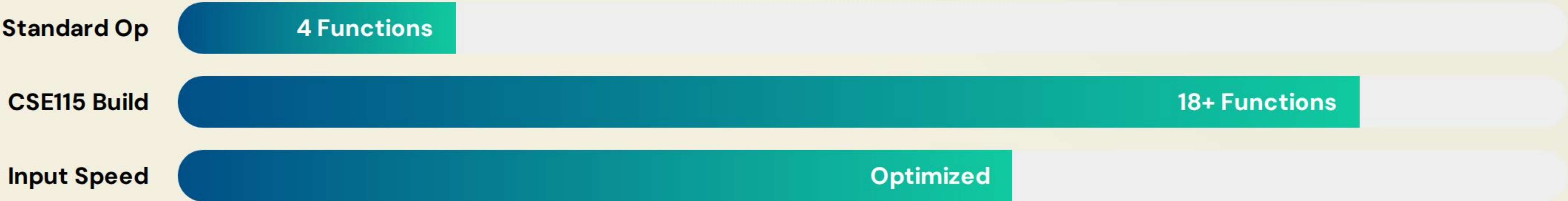
Source Implementation



Modular Programming

Functions are isolated into dedicated blocks: `add()`, `subtract()`, `sinFunc()`, etc. This enhances readability for the CSE115 evaluation.

Capabilities Comparison



Comparison of standard calculator builds vs. our comprehensive scientific implementation.

Interactive CLI

```
--- Scientific Calc ---
```

```
1. Addition
```

```
2. Sin(x)
```

```
3. Exit
```

```
Enter Choice: 2
```

```
Enter Degrees: 90
```

```
Result: 1.0000
```

Continuous Operation

The program utilizes do-while loops to allow users to perform multiple calculations without restarting the binary, providing a seamless user experience.

Thank You

Faculty: MSRB | Course: CSE115

Team Members:

Shafayet Arefin • Sabbir Ahmed

Samin Yasar Ahmed • Samiul Islam Sami

Q & A

Any Questions?

Image Sources



https://ae-pic-a1.aliexpress-media.com/kf/S54bb762436824b4fa18c8159ebb5e469T.jpg?has_lang=1&ver=2_960x960q75.jpg_.avif

Source: www.aliexpress.com



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